

FALL 2026 | MEE4033

Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 6

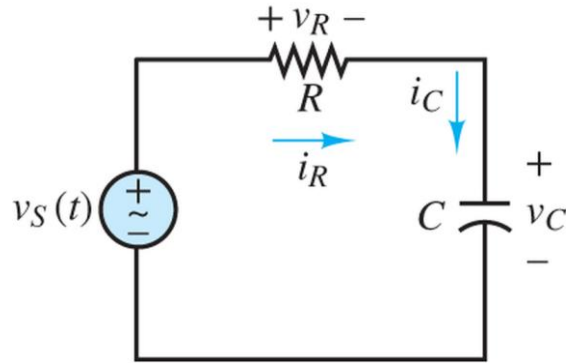
Transient Analysis

Principles and Applications of Electrical Engineering, 7th ed., Chapter 4
Electric Circuits, 11th ed., Chapters 7, 8 and 9

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4.1 Transient Analysis: Where the Transient Comes From

- AC circuit



(1st-order ODE)

$$\frac{dx}{dt} + x = V_s$$

$$\frac{dx_N}{dt} + x_N = 0$$

$$\frac{dx_F}{dt} + x_F = V_s$$

Solution has two parts:

- A **transient solution** (homogeneous, natural response)
- A **steady-state solution** (particular, forced response)

Complete response = **transient (natural) response** + **steady-state (forced) response**:

$$\mathbf{x} = \mathbf{x}_N + \mathbf{x}_F$$

4.1 Transient Analysis: Switching

Switching

- Example process that produces a transient response.
 - Abrupt change between short-circuit $\leftrightarrow$ open-circuit
 - Eventually declines to zero and only the steady-state solution remains (to a DC or AC steady-state solution).
 - 'Transition' to a new state

Important aspects of a transient event

- Initial conditions on the state variables before the event.
- The manner and speed of the transition (due to energy storage in capacitors and inductors).

4.1 Transient Analysis: State Variables

State variables

- The energy stored in capacitors and inductors is a function of the capacitor voltage and the inductor current.
 - Capacitor: $w_C = (1/2) C v_C^2$, so v_C is the state variable.
 - Inductor: $w_L = (1/2) L i_L^2$, so i_L is the state variable.
 - A state variable cannot jump: an instantaneous change would require infinite power.

4.1 Transient Analysis: First- and Second-Order Circuits

First-order circuit (RC, RL)

- Contains a single storage element.

Second-order circuit (series/parallel LC)

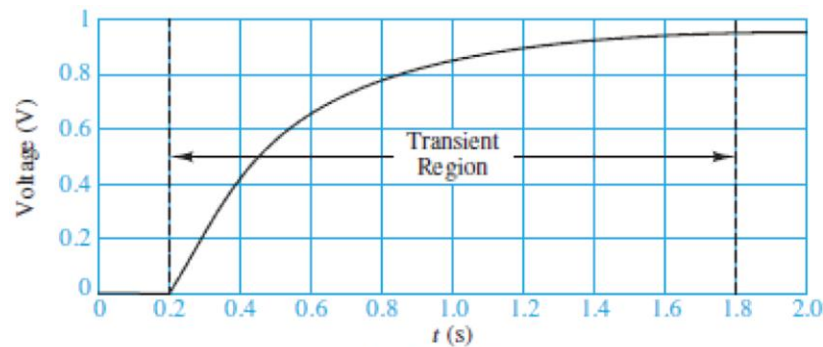
- Contains two irreducible storage elements.

Objective of transient analysis

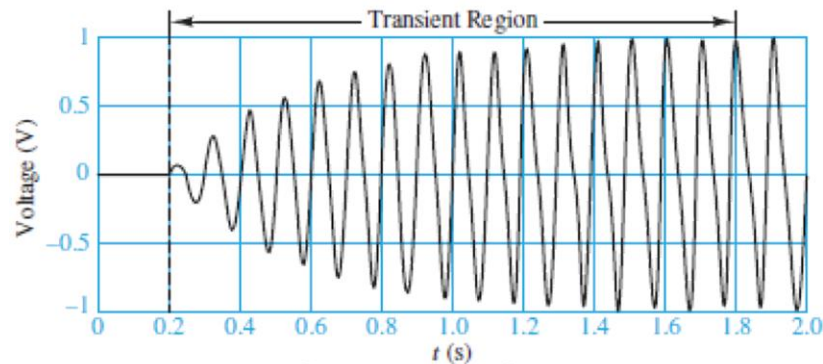
- Determine the manner and speed with which voltages and currents transition from one steady state to another.

First- and Second-Order Transient Responses

- Initial steady state → transient response → final steady state (Rizzoni Figure 4.1)



(a) Transient DC voltage



(b) Transient sinusoidal voltage

Figure 4.1 First- and second-order transient responses

Switching a 12 V source onto a parallel LC load: (a) first-order-like DC transient, (b) second-order transient with oscillation.

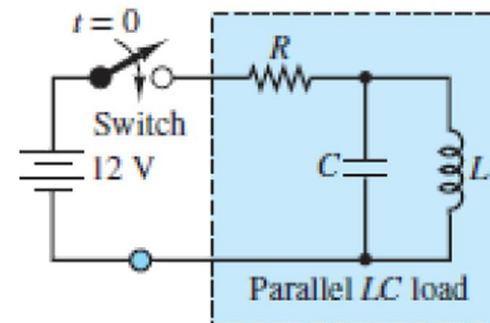


Figure 4.2 Circuit with switched DC excitation

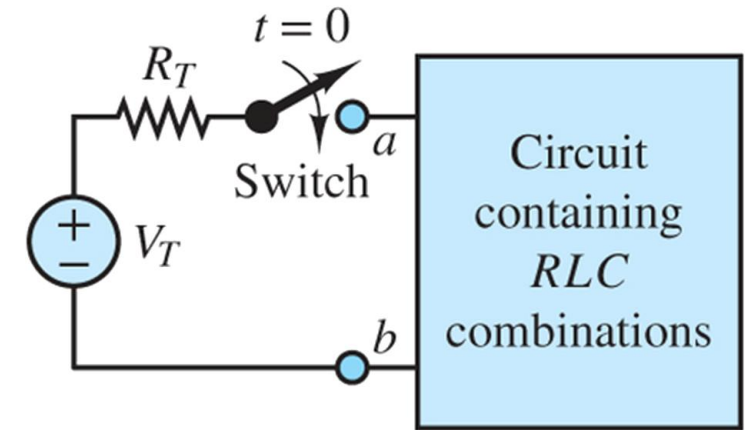
General Models of the Transient Analysis Problem

Transient response of a first-order load

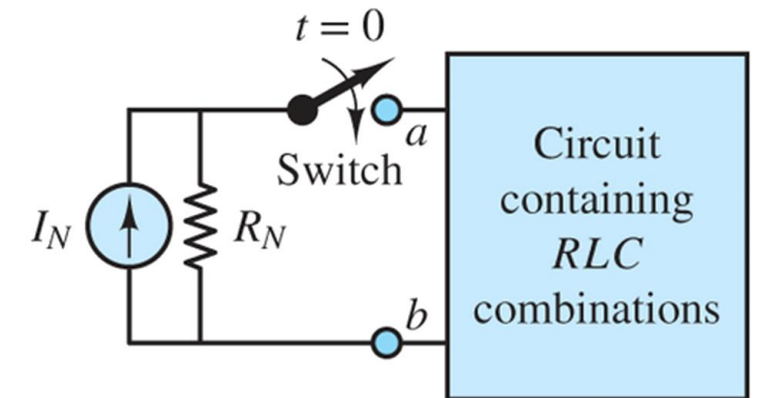
- Rising or falling exponential.

Response of second-order circuits depends on the damping ratio

- Overdamped.
- Underdamped.
- Critically damped.



(a)

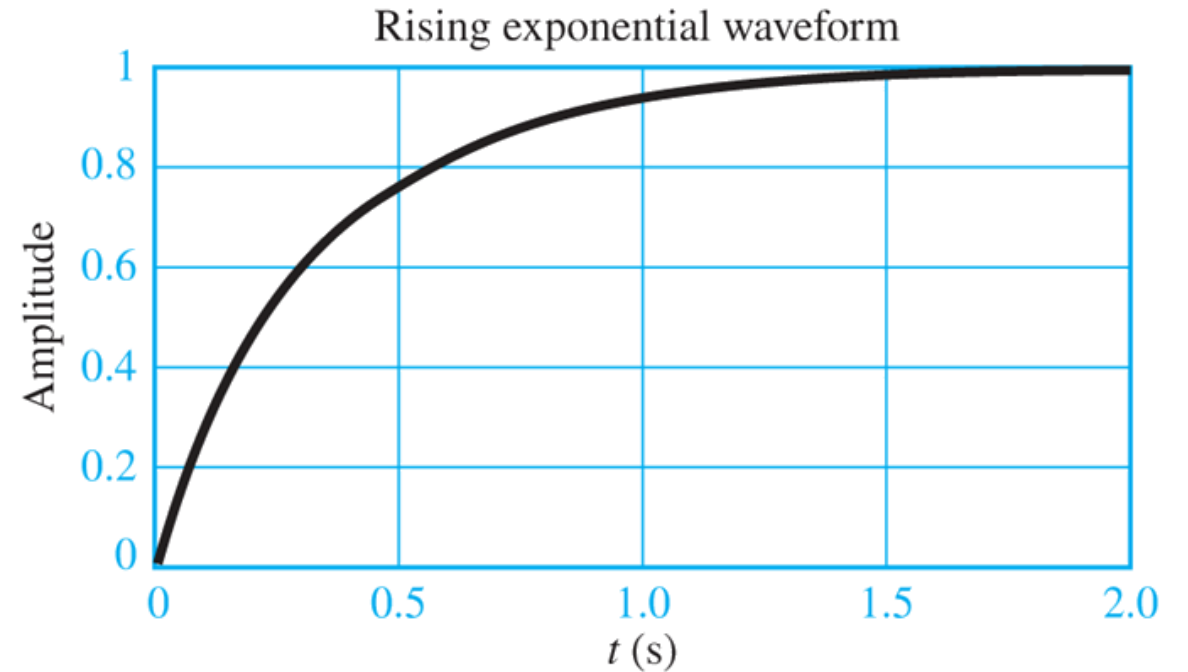
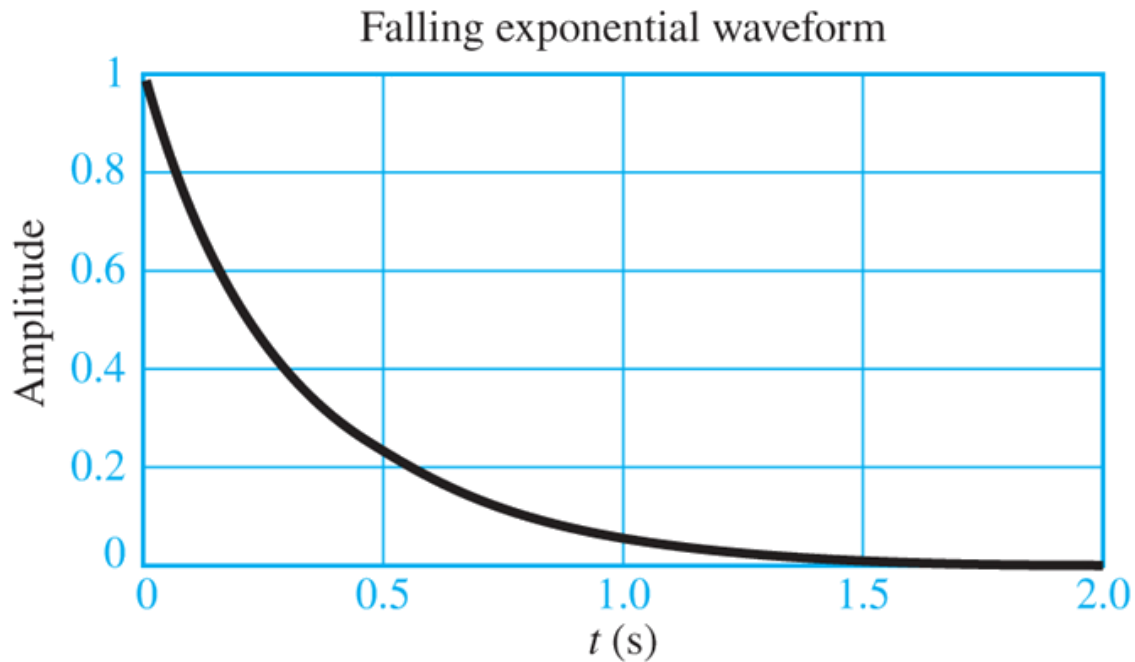


(b)

Falling and Rising Exponential Responses

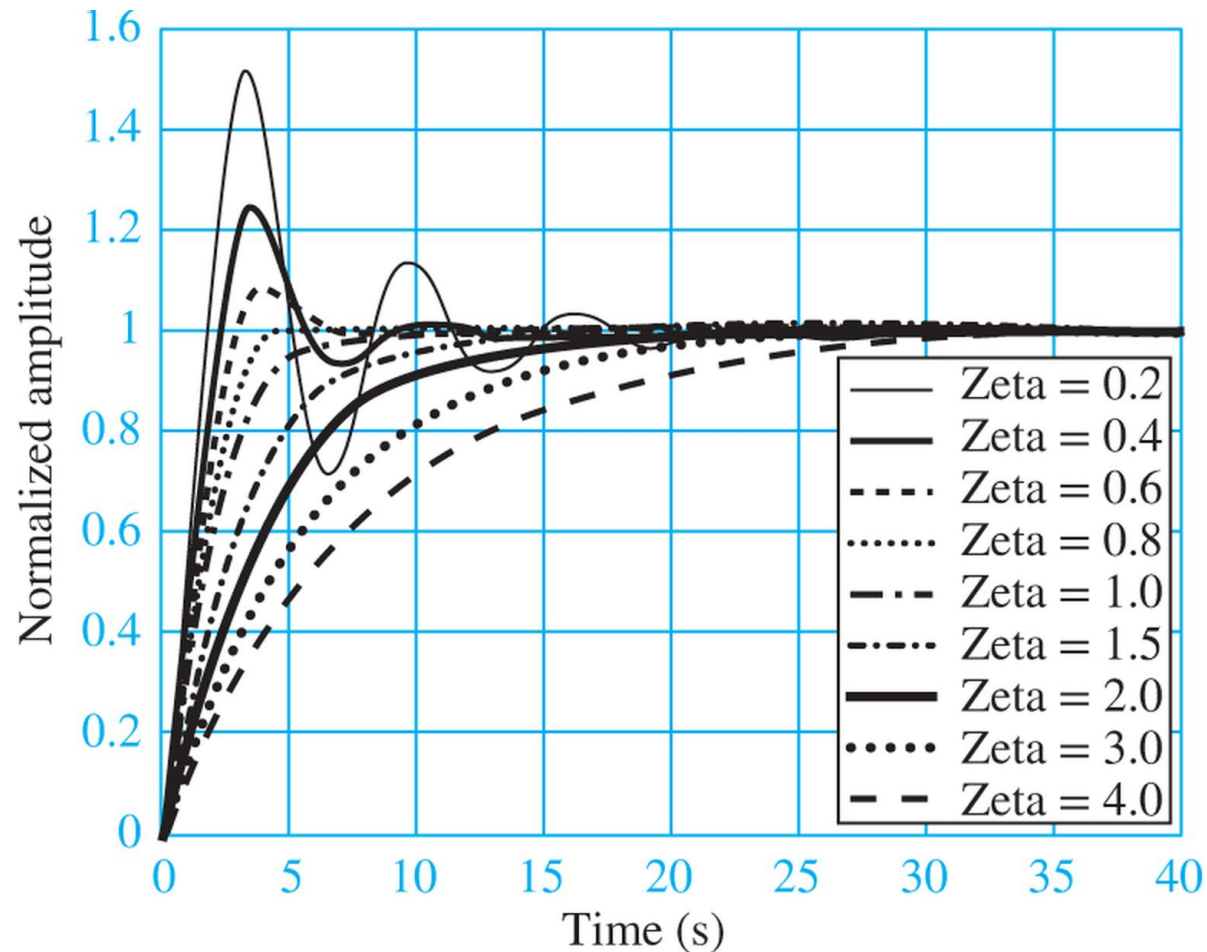
First-order circuit (RC, RL)

- Rising or falling exponential



Second-Order Transient Responses

Second-order circuit (series/parallel LC)



- Response of second-order circuits depends on the damping ratio ζ .
 - Overdamped ($\zeta > 1$)
 - Underdamped ($\zeta < 1$)
 - Critically damped ($\zeta = 1$)

4.2 Elements of Transient Problem Solving

Moment of a transient event known as $t = 0$.

Initial steady state: $t < 0$.

- In a DC steady state, a capacitor acts as an open and an inductor acts as a short.

State variables

- Currents through inductors and voltages across capacitors.

- Always continuous functions of time.

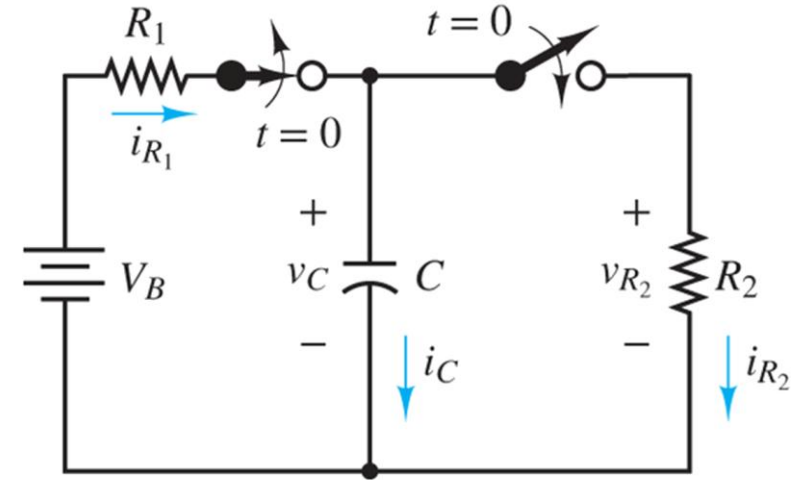
$$i_C = C \frac{dv_C}{dt} \quad \text{and} \quad v_L = L \frac{di_L}{dt}$$

- v_C and i_L are continuous
- Non-state variables may not be continuous.

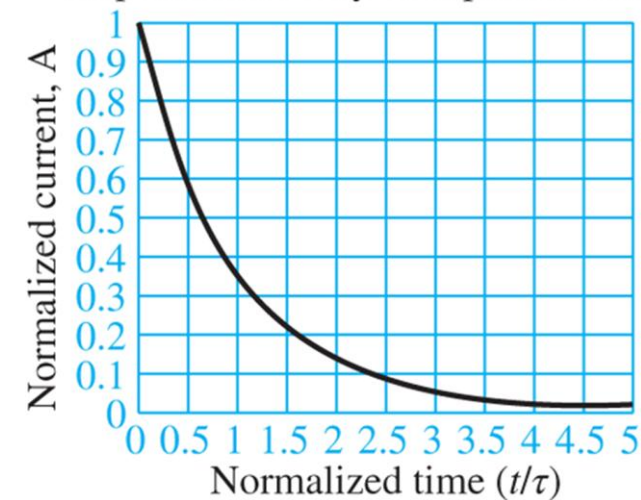
Energy and the Transient Response

- Energy is supplied, exchanged, or dissipated until steady state is reached.
- KCL: $i_C + i_{R_2} = 0$
 - $v_{R_2} = i_{R_2} R_2$
 - $i_C = C \frac{dv_C}{dt}$

$$\frac{dv_C}{dt} = -\frac{1}{R_2 C} v_C$$



Exponential decay of capacitor current

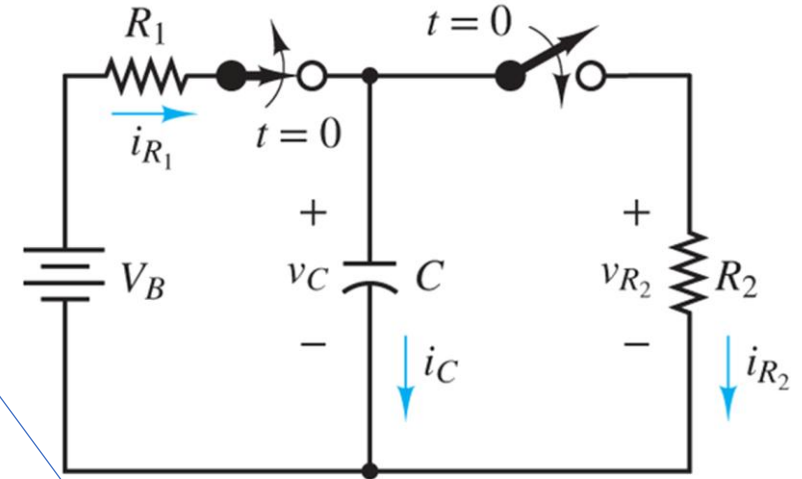


Energy and the Transient Response

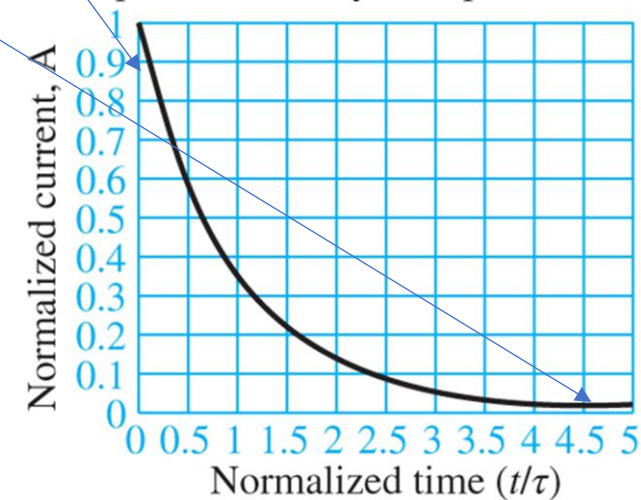
At $t = 0+$ the capacitor is discharging at its maximum rate, because v_C itself and, thus, dv_C/dt are both at their maximum at that moment.

As the capacitor continues to discharge, v_C and dv_C/dt continue to decrease, so the rate of decrease of v_C decreases as well.

$$\frac{dv_C}{dt} = -\frac{1}{R_2 C} v_C$$



Exponential decay of capacitor current



Transient Responses: Long-Term Steady State and Complete Response

Long-term steady state

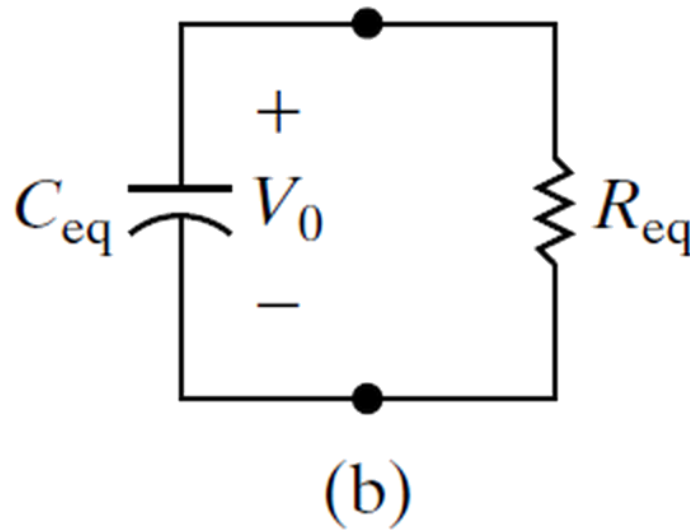
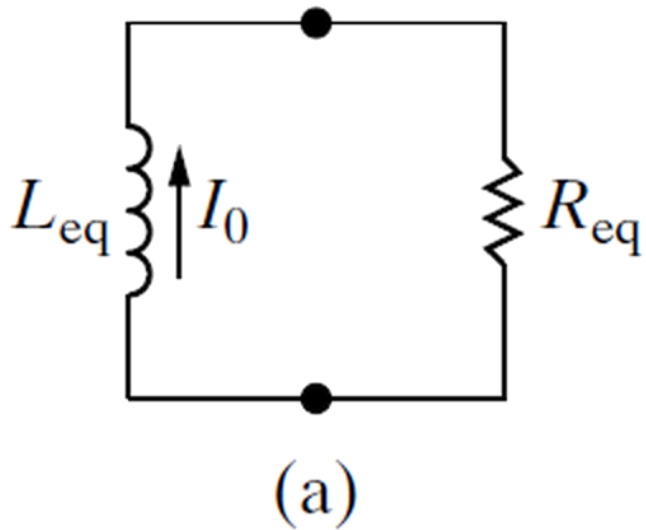
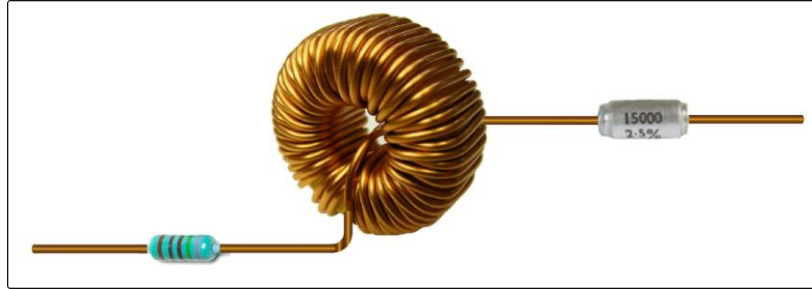
- Depends on the independent sources present in the $t > 0$ circuit.
- Expressed as a gain K multiplied by a forcing function $F(t)$.

Complete response

- Sum of the transient and long-term steady-state responses.
 - The two components are also called the natural and forced responses.

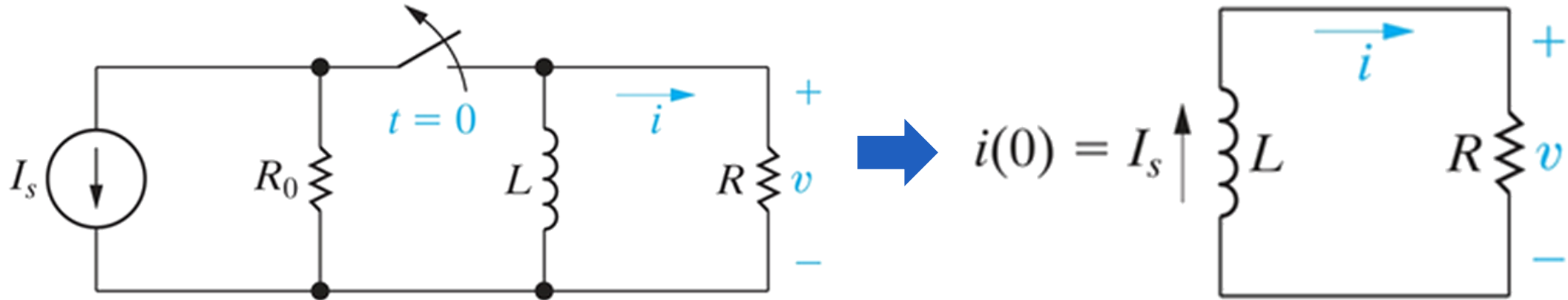
4.3 First-Order Transient Analysis

- RC or RL circuit (= mass-damper system in a mechanical system)



- (a) RL (resistor-inductor) circuit
- (b) RC (resistor-capacitor) circuit

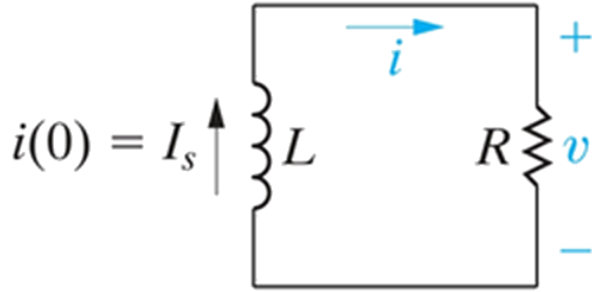
RL Natural Response



The problem: Given an initial current $i(0)$ through L at $t = 0$, find $i(t)$ for $t \geq 0$.

- Note: this is the 'natural' response of this circuit because for $t \geq 0$ there is no independent source in the circuit. The response is due only to the 'nature' of the components and their interconnection.

RL Natural Response: The Describing Equation

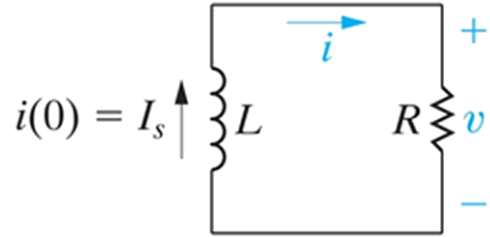


KVL :
$$L \frac{di(t)}{dt} + Ri(t) = 0$$

This equation is a

- First-order (the highest derivative is a first derivative)
- Homogeneous (the right-hand side is 0)
- Ordinary differential equation
- With constant coefficients

RL Natural Response: Solution 1 (Separation of Variables)



$$\text{KVL : } L \frac{di(t)}{dt} + Ri(t) = 0$$

- Natural response

$$\frac{di}{dt} dt = -\frac{R}{L} i dt$$

$$\frac{di}{i} = -\frac{R}{L} dt$$

$$\int_{i(t_0)}^{i(t)} \frac{dx}{x} = -\frac{R}{L} \int_{t_0}^t dy$$

$$\ln \frac{i(t)}{i(0)} = -\frac{R}{L} t$$

$$i(t) = i(0)e^{-(R/L)t}$$

- Power dissipated in R

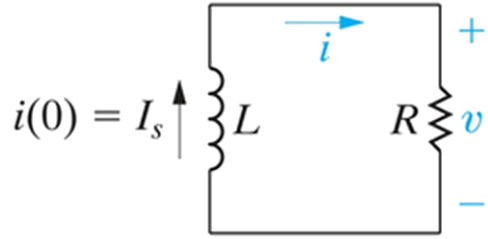
$$w = \int_0^t p dx = \int_0^t I_0^2 R e^{-2x/\tau} dx$$

$$= \frac{\tau}{2} I_0^2 R (1 - e^{-2t/\tau})$$

$$= \frac{L}{2R} I_0^2 R (1 - e^{-2t/\tau})$$

$$= \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau}), \quad t \geq 0^+$$

RL Natural Response: Solution 2 (Assume an Exponential)



$$\text{KVL :} \quad L \frac{di(t)}{dt} + Ri(t) = 0$$

$$\text{Let } i(t) = e^{-at} \quad \text{and} \quad \frac{di(t)}{dt} = -ae^{-at}$$

$$\text{Substituting,} \quad L(-ae^{-at}) + Re^{-at} = 0$$

$$\text{Rearranging,} \quad (R - aL)e^{-at} = 0 \quad \text{so} \quad R - aL = 0$$

$$\text{Solving,} \quad a = R/L$$

$$\text{But } i(0) = I_s \quad \text{so} \quad i(t) = I_s e^{-(R/L)t}, \quad t \geq 0$$

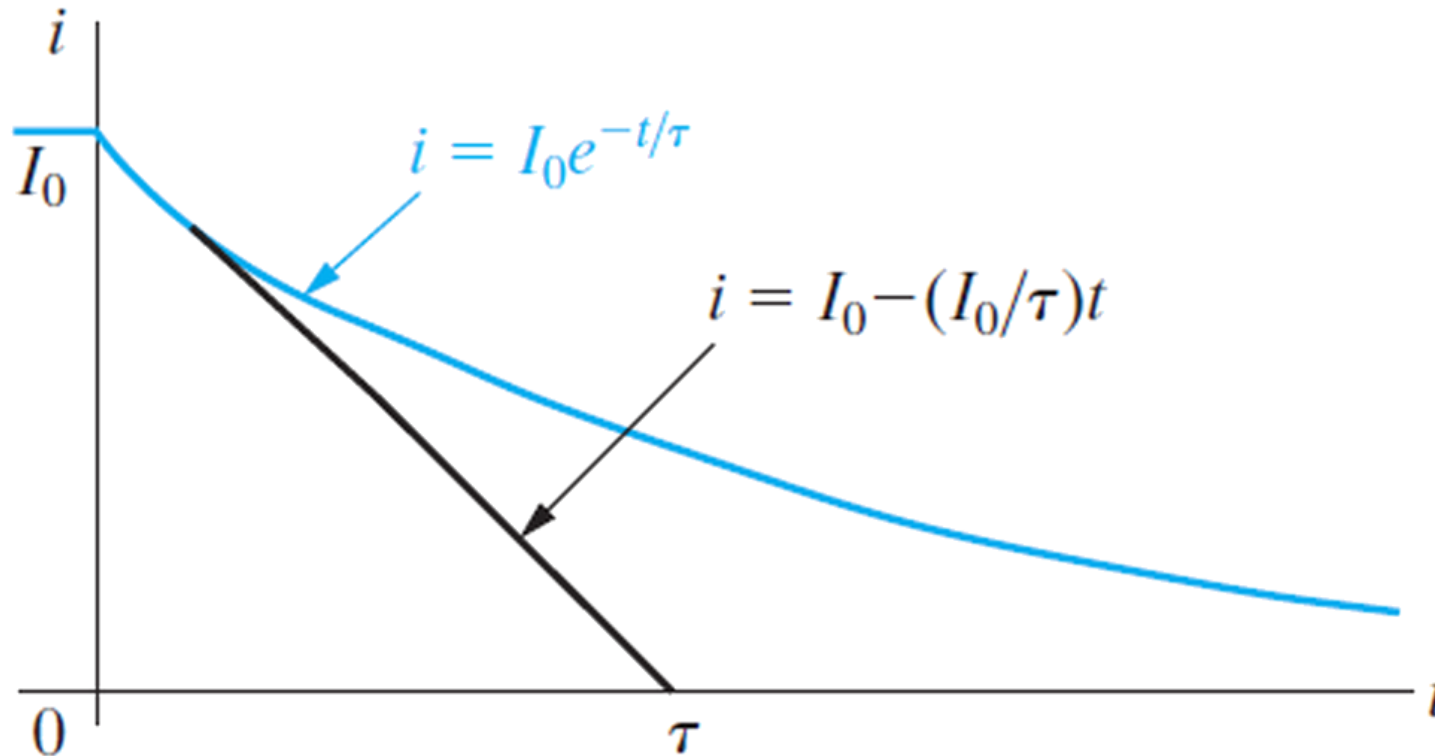
$$\begin{aligned} \text{Check:} \quad & L(-R/L)I_s e^{-(R/L)t} + RI_s e^{-(R/L)t} \\ & = -RI_s e^{-(R/L)t} + RI_s e^{-(R/L)t} = 0 \end{aligned}$$

$$i(0) = I_s e^{-(R/L)(0)} = I_s$$

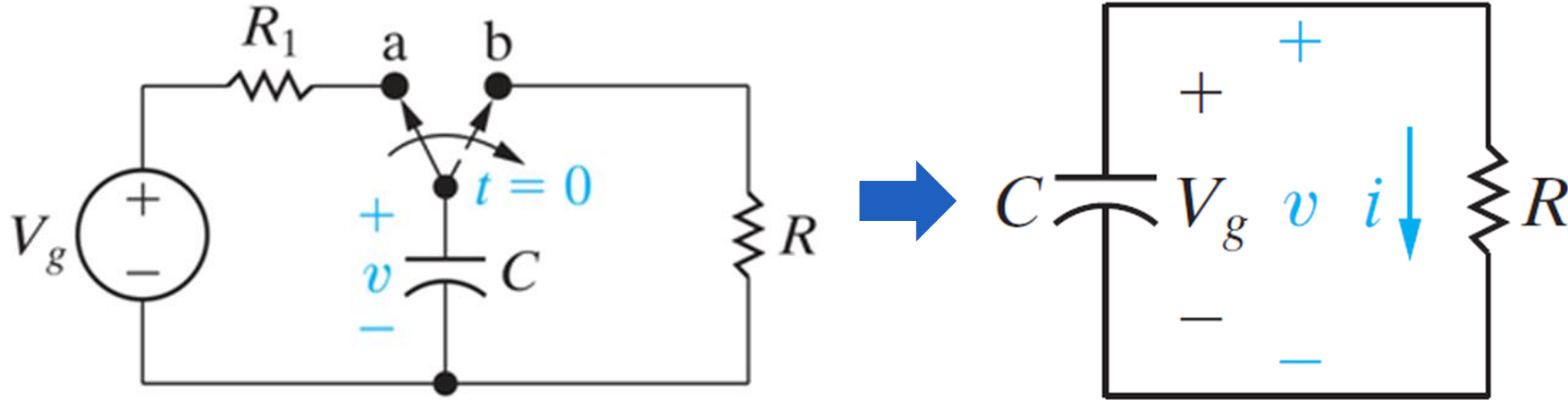
RL Natural Response: Time Constant

TIME CONSTANT FOR *RL* CIRCUIT

$$\tau = \frac{L}{R}.$$



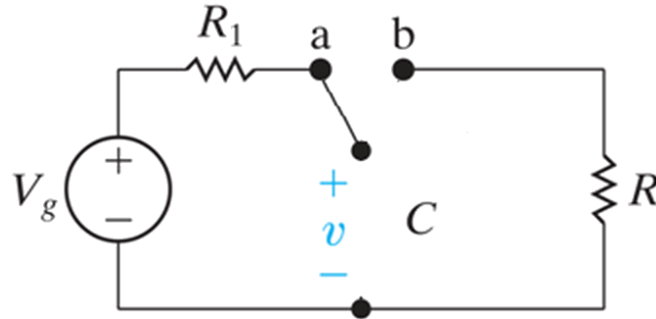
RC Natural Response



The problem: Find $v(t)$ for $t \geq 0$. Note that the circuit to the left of the capacitor establishes an initial voltage drop $v(0)$ across C at $t = 0$.

RC Natural Response: Initial Condition

$t < 0$:



There is an open circuit here, so the current in this circuit is 0, and the voltage drop across R_1 is 0. Thus,

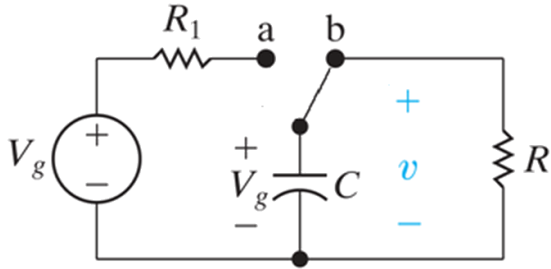
$$v = V_g, \quad t < 0$$

Remember the continuity requirement for capacitors: the voltage drop across the capacitor must be continuous everywhere, so that the current through the capacitor remains finite. Thus,

$$v(0) = V_g$$

RC Natural Response: Solution 1

$t \geq 0 :$



- Natural response

Derive the equation yourself!

$$v(t) = v(0)e^{-t/RC}$$

KCL:
$$C \frac{dv(t)}{dt} + \frac{v(t)}{R} = 0$$

- Power dissipated in R

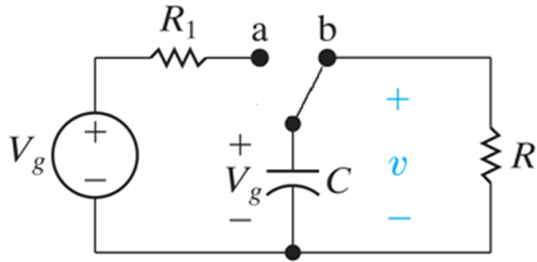
$$i(t) = \frac{v(t)}{R} = \frac{V_0}{R} e^{-t/\tau}, \quad t \geq 0^+,$$

$$p = vi = \frac{V_0^2}{R} e^{-2t/\tau}, \quad t \geq 0^+,$$

$$\begin{aligned} w &= \int_0^t p \, dx = \int_0^t \frac{V_0^2}{R} e^{-2x/\tau} \, dx \\ &= \frac{1}{2} CV_0^2 (1 - e^{-2t/\tau}), \quad t \geq 0. \end{aligned}$$

RC Natural Response: Solution 2

$t \geq 0$:



KCL:
$$C \frac{dv(t)}{dt} + \frac{v(t)}{R} = 0$$

Let $v(t) = e^{-at}$ and $\frac{dv(t)}{dt} = -ae^{-at}$

Substituting, $C(-ae^{-at}) + e^{-at}/R = 0$

Rearranging, $(1/R - aC)e^{-at} = 0$ so $1/R - aC = 0$

Solving, $a = 1/RC$

But $v(0) = V_g$ so $v(t) = V_g e^{-(1/RC)t}, \quad t \geq 0$

RC Natural Response: Time Constant

TIME CONSTANT FOR RC CIRCUIT

$$\tau = RC.$$

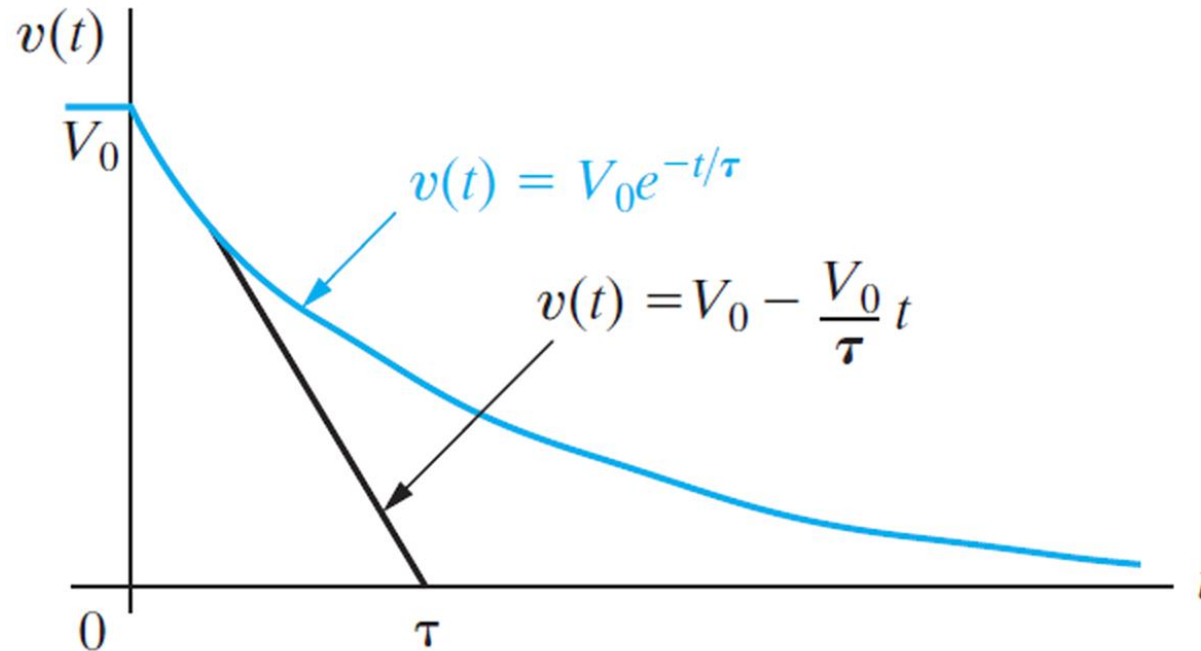
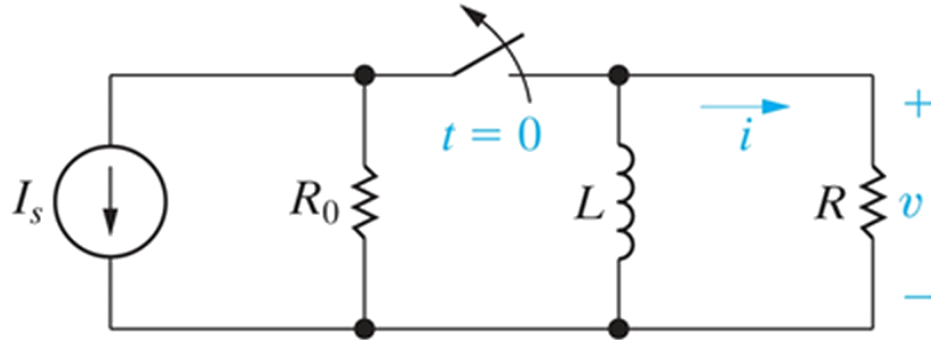


Figure 7.14 ▲ The natural response of an RC circuit.

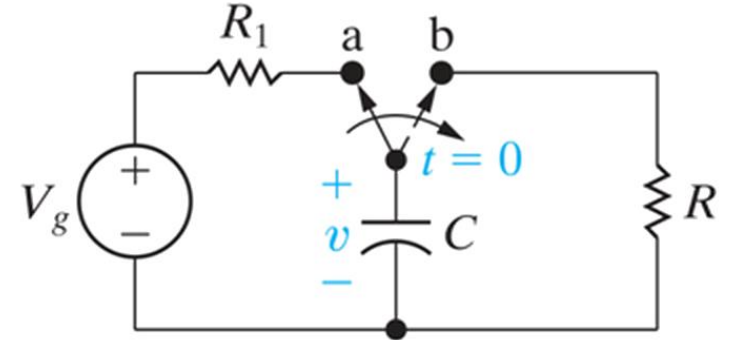
RL/RC Natural Response: Summary



$$\text{KVL:} \quad L \frac{di(t)}{dt} + Ri(t) = 0$$

$$i(0) = I_s$$

$$i(t) = I_s e^{-(R/L)t}, \quad t \geq 0$$



$$\text{KCL:} \quad C \frac{dv(t)}{dt} + \frac{v(t)}{R} = 0$$

$$v(0) = V_g$$

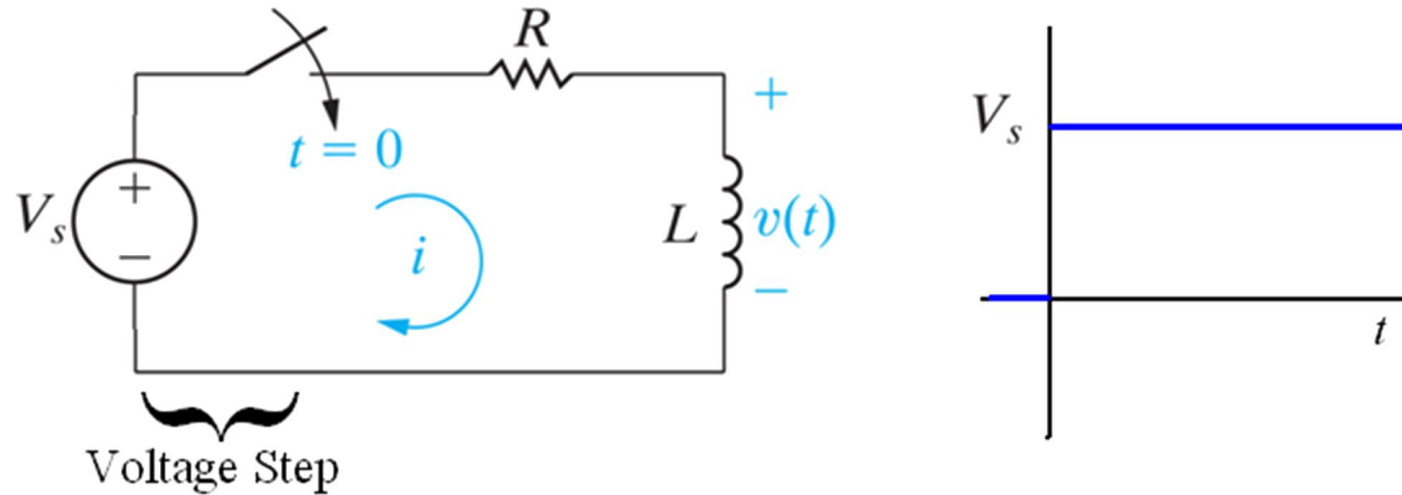
$$v(t) = V_g e^{-(1/RC)t}, \quad t \geq 0$$

- The form of the natural response is the same: $\mathbf{IC}e^{-t/\tau}$

IC is the initial condition and τ is the time constant, a measure of how quickly or slowly the exponential decays.

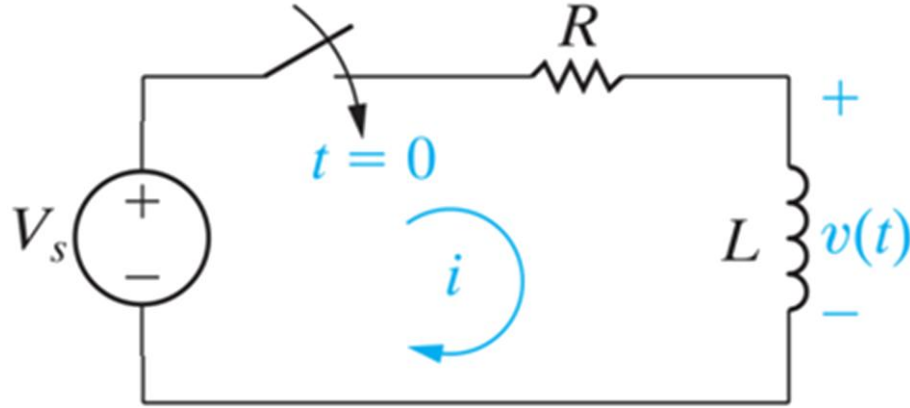
RL Step Response

The problem: Given an initial current $i(0)$ through L at $t = 0$, find $i(t)$ for $t \geq 0$. Note that the initial current may be 0!



This is the step response of this circuit because for $t \geq 0$ there is an independent source in the circuit. The response is due to the initial energy stored in the inductor as well as the energy supplied by the voltage source for $t \geq 0$.

RL Step Response: The Describing Equation



- KVL for $t \geq 0$:

$$-V_s + L \frac{di(t)}{dt} + Ri(t) = 0$$

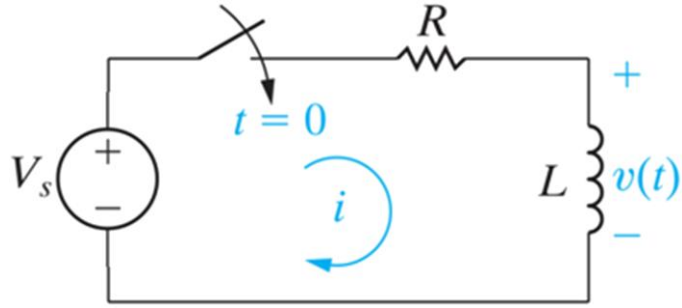
$$\Rightarrow L \frac{di(t)}{dt} + Ri(t) = V_s$$

This equation is a

- First-order, inhomogeneous (the right-hand side is non-zero), ordinary differential equation with constant coefficients.

The left-hand side of the equation is identical to the describing differential equation for the RL natural response!

RL Step Response: Solution 1 (Separation of Variables)



KVL :
$$L \frac{di(t)}{dt} + Ri(t) = V_s$$

$$\frac{di}{dt} = \frac{-Ri + V_s}{L} = \frac{-R}{L} \left(i - \frac{V_s}{R} \right)$$

$$di = \frac{-R}{L} \left(i - \frac{V_s}{R} \right) dt$$

$$\frac{di}{i - (V_s/R)} = \frac{-R}{L} dt$$

$$\int_{I_0}^{i(t)} \frac{dx}{x - (V_s/R)} = \frac{-R}{L} \int_0^t dy$$

$$\ln \frac{i(t) - (V_s/R)}{I_0 - (V_s/R)} = \frac{-R}{L} t$$

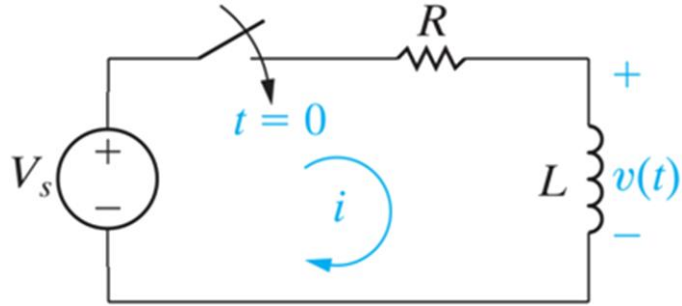
$$\frac{i(t) - (V_s/R)}{I_0 - (V_s/R)} = e^{-(R/L)t}$$

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-(R/L)t}$$

STEP RESPONSE OF AN RL CIRCUIT

$$i(t) = I_f + (I_0 - I_f) e^{-t/\tau}$$

RL Step Response: Solution 2 (Guess and Check)



$$\text{KVL :} \quad L \frac{di(t)}{dt} + Ri(t) = V_s$$

The functional form for $i(t)$ should be the same as for the natural response!

Try $i(t) = I_0 e^{-(R/L)t}$, where I_0 is the initial inductor current.

But as $t \rightarrow \infty$ $i(\infty) = I_0 e^{-(R/L)(\infty)} = 0$ which is incorrect!

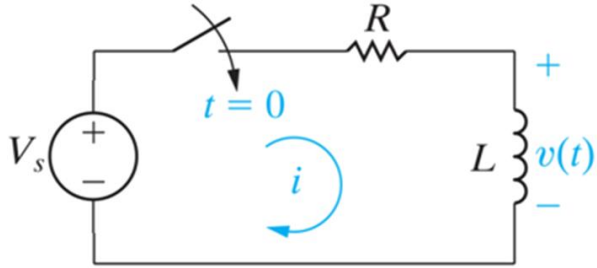
Try $i(t) = I_F + I_0 e^{-(R/L)t}$,

where I_F is the inductor current as $t \rightarrow \infty$.

Then $i(\infty) = I_F + I_0 e^{-(R/L)(\infty)} = I_F$ (OK)

But $i(0) = I_F + I_0 e^{-(R/L)(0)} = I_F + I_0$ (NO!)

RL Step Response: Solution 2 (continued)



$$\text{KVL :} \quad L \frac{di(t)}{dt} + Ri(t) = V_s$$

Now try $i(t) = I_F + (I_0 - I_F)e^{-(R/L)t}$

Then $i(\infty) = I_F + (I_0 - I_F)e^{-(R/L)(\infty)} = I_F$ (OK)

And $i(0) = I_F + (I_0 - I_F)e^{-(R/L)(0)} = I_F + I_0 - I_F = I_0$ (OK)

Make sure that this $i(t)$ satisfies the differential equation:

$$i(t) = I_F + (I_0 - I_F)e^{-(R/L)t}; \quad \frac{di(t)}{dt} = -(R/L)(I_0 - I_F)e^{-(R/L)t}$$

Substituting,

$$L[-(R/L)(I_0 - I_F)e^{-(R/L)t}] + R[I_F + (I_0 - I_F)e^{-(R/L)t}] = RI_F = V_s$$

RL Step Response: Current and Inductor Voltage

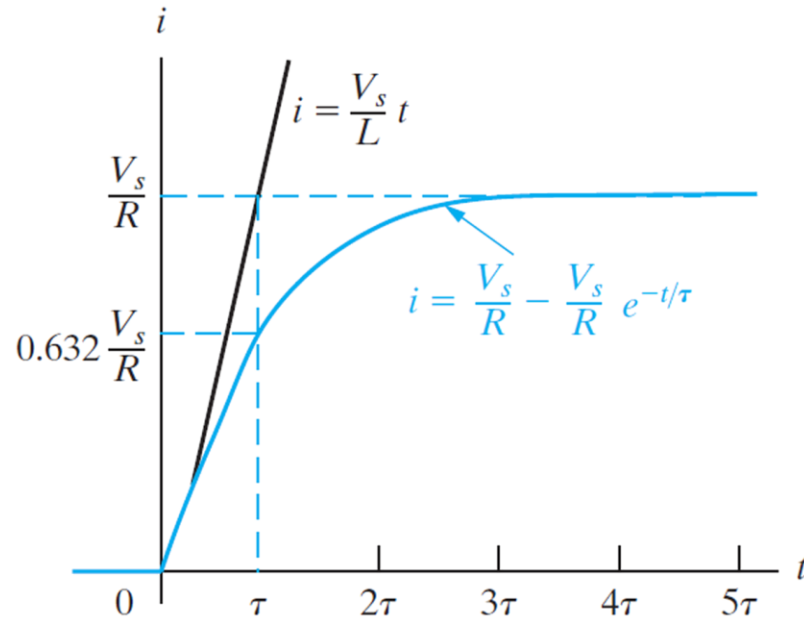


Figure 7.26 ▲ The step response of the RL circuit shown in Fig. 7.20 when $I_0 = 0$.

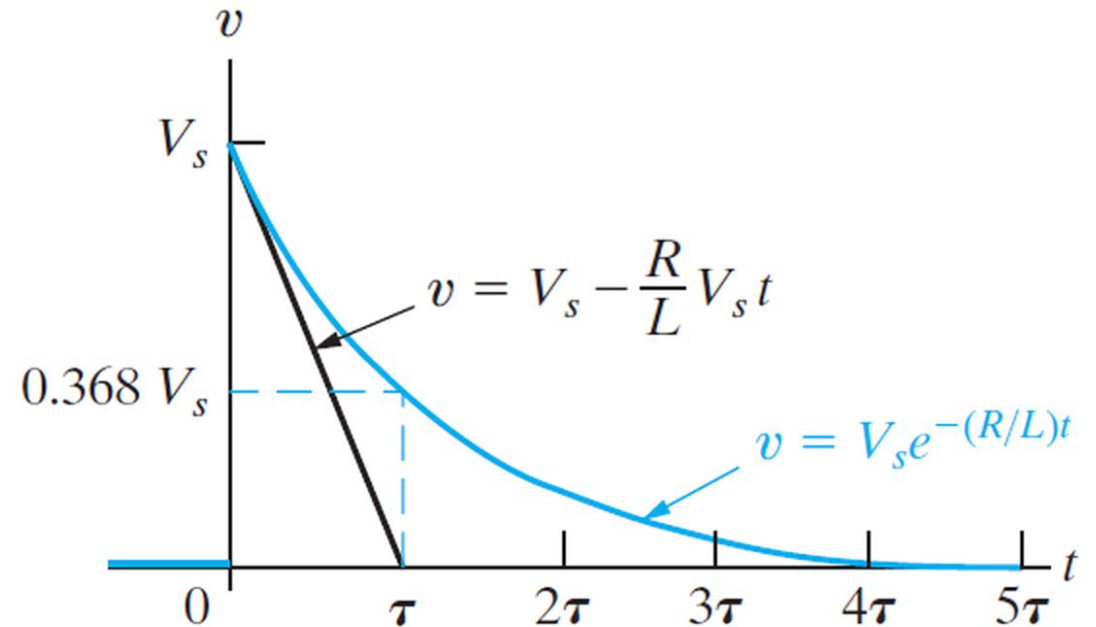
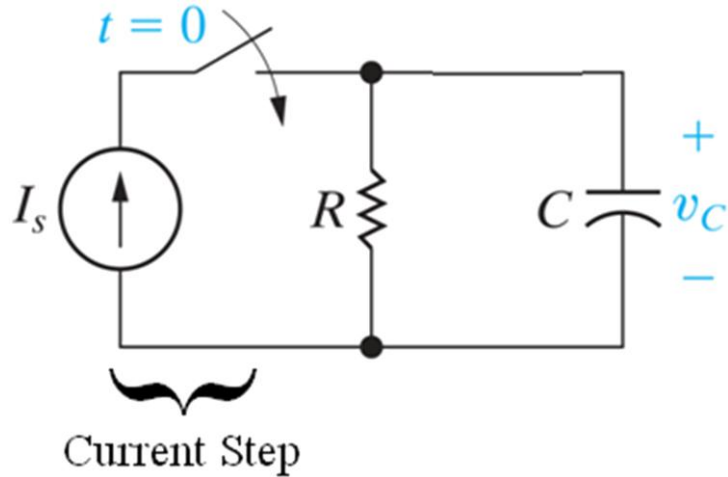


Figure 7.27 ▲ Inductor voltage versus time.

- As $t \rightarrow \infty$ the inductor becomes a short circuit, so $i \rightarrow V_s/R$ and $v_L \rightarrow 0$.
 - Nilsson Figures 7.26 and 7.27, drawn for $I_0 = 0$.

RC Step Response



- KCL: $C \frac{dv}{dt} + \frac{v}{R} = I_s$

The problem: Given an initial voltage drop $v(0)$ across C at $t = 0$, find $v(t)$ for $t \geq 0$. Note that the initial voltage may be 0!

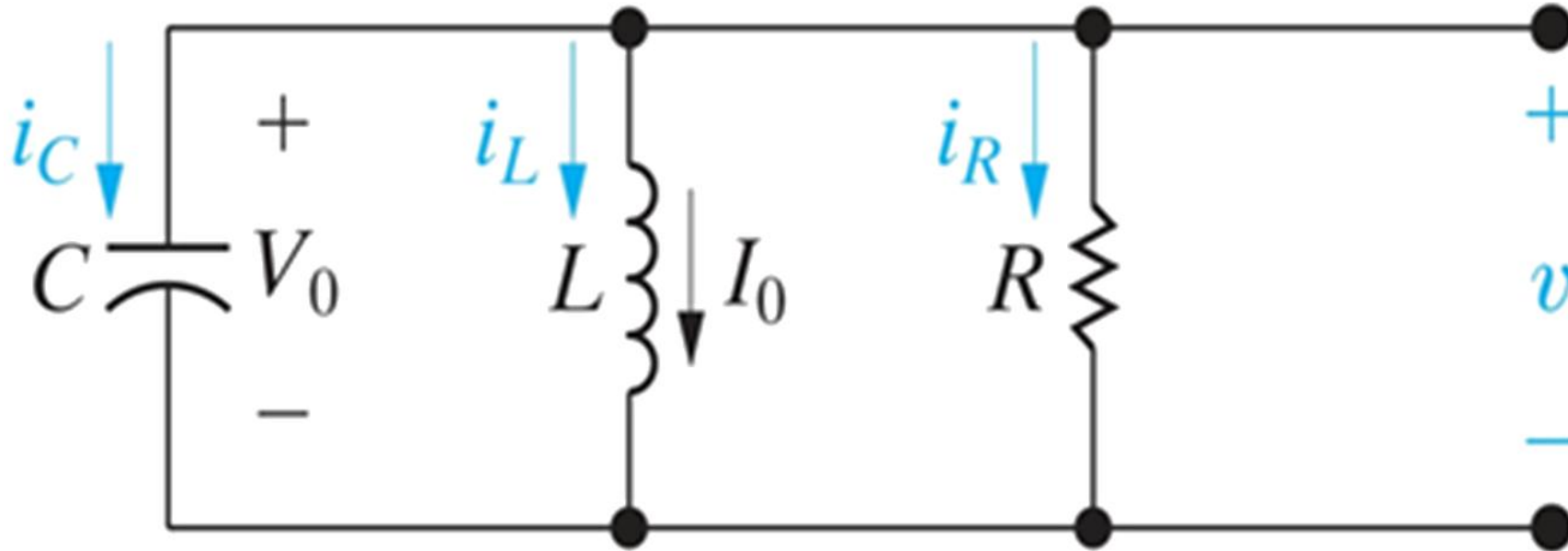
This is the dual of the RL step response problem, so its solution is the dual of the solution to the RL step response.

$$v_C(t) = V_F + (V_0 - V_F)e^{-t/RC}, \quad t \geq 0$$

where $V_0 = v_C(0)$ and $V_F = v_C(\infty)$

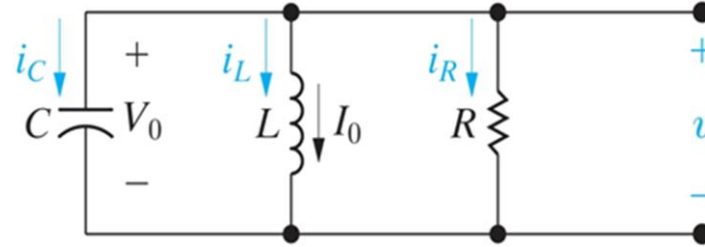
4.4 Second-Order Transient Analysis: Parallel RLC Natural Response

The problem: given initial energy stored in the inductor and/or capacitor, find $v(t)$ for $t \geq 0$.



Parallel RLC Natural Response: The Describing Equation

The problem: given initial energy stored in the inductor and/or capacitor, find $v(t)$ for $t \geq 0$.



1. KCL at the top node:

$$i_C + i_L + i_R = 0$$

2. Element laws:

$$i_C = C \frac{dv}{dt}, \quad i_R = \frac{v}{R}, \quad i_L = \frac{1}{L} \int_0^t v d\tau + I_0$$

3. Substitute into KCL:

$$C \frac{dv}{dt} + \frac{1}{L} \int_0^t v d\tau + I_0 + \frac{v}{R} = 0$$

4. Differentiate (I_0 is constant):

$$C \frac{d^2v}{dt^2} + \frac{1}{L} v + \frac{1}{R} \frac{dv}{dt} = 0$$

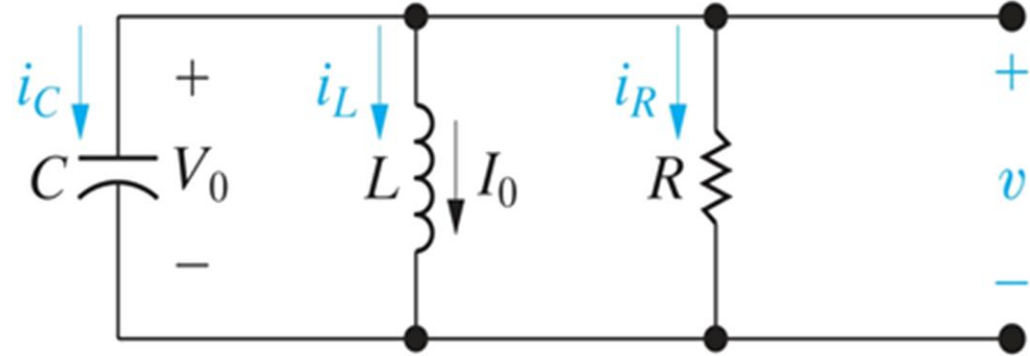
5. Divide by C:

$$\frac{d^2v}{dt^2} + \frac{1}{LC} v + \frac{1}{RC} \frac{dv}{dt} = 0$$

Second-order, homogeneous, ordinary differential equation with constant coefficients.

Parallel RLC Natural Response: Characteristic Equation

The problem: given initial energy stored in the inductor and/or capacitor, find $v(t)$ for $t \geq 0$.



Describing equation:
$$\frac{d^2v(t)}{dt^2} + \frac{1}{LC}v(t) + \frac{1}{RC}\frac{dv(t)}{dt} = 0$$

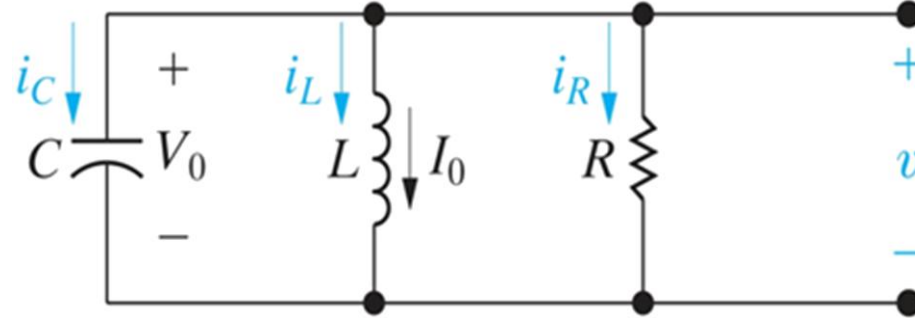
Solution :
$$v(t) = A_1e^{s_1t} + A_2e^{s_2t}$$

Both s_1 and s_2 are solutions of the CHARACTERISTIC EQUATION

$$s^2 + (1/RC)s + (1/LC) = 0$$

Parallel RLC Natural Response: Two Constants

The problem: given initial energy stored in the inductor and/or capacitor, find $v(t)$ for $t \geq 0$.



Describing equation:
$$\frac{d^2v(t)}{dt^2} + \frac{1}{LC}v(t) + \frac{1}{RC}\frac{dv(t)}{dt} = 0$$

The circuit has two initial conditions that must be satisfied, so the solution for $v(t)$ must have two constants. Use

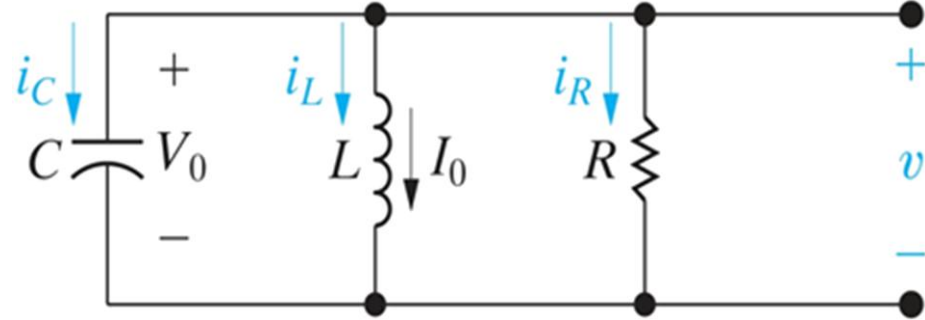
$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \text{ V}; \quad \text{Substitute:}$$

$$(s_1^2 A_1 e^{s_1 t} + s_2^2 A_2 e^{s_2 t}) + \frac{1}{RC}(s_1 A_1 e^{s_1 t} + s_2 A_2 e^{s_2 t}) + \frac{1}{LC}(A_1 e^{s_1 t} + A_2 e^{s_2 t}) = 0$$

$$\Rightarrow [s_1^2 + (1/RC)s_1 + (1/LC)]A_1 e^{s_1 t} + [s_2^2 + (1/RC)s_2 + (1/LC)]A_2 e^{s_2 t} = 0$$

Parallel RLC Natural Response: Roots of the Characteristic Equation

The problem: given initial energy stored in the inductor and/or capacitor, find $v(t)$ for $t \geq 0$.



The two solutions to the characteristic equation can be calculated using the quadratic formula:

$$s^2 + (1/RC)s + (1/LC) = 0; \quad s_{1,2} = \frac{-(1/RC) \pm \sqrt{(1/RC)^2 - 4(1/LC)}}{2}$$

$$s_{1,2} = -(1/2RC) \pm \sqrt{(1/2RC)^2 - (1/LC)} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

where $\alpha = \frac{1}{2RC}$ (the neper frequency in rad/s)

and $\omega_0 = \sqrt{\frac{1}{LC}}$ (the resonant radian frequency in rad/s)

Neper frequency α (underdamped case)

$$v(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

α is the decay rate of the envelope $e^{-\alpha t}$: after $1/\alpha$ seconds it falls to $1/e$ ($\approx 37\%$).

Unit: Np/s (dimension 1/s).

Parallel RLC Natural Response: Three Forms of the Roots

So far, we know that the parallel RLC natural response is given by

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \quad \text{where} \quad \alpha = \frac{1}{2RC} \quad \text{and} \quad \omega_0 = \sqrt{\frac{1}{LC}}$$

There are three different forms for s_1 and s_2 . For a parallel RLC circuit with specific values of R, L and C, the form for s_1 and s_2 depends on

-  A. The value of α
-  B. The value of ω_0
-  C. The value of $(\alpha^2 - \omega_0^2)$

Parallel RLC: Overdamped, Underdamped, Critically Damped

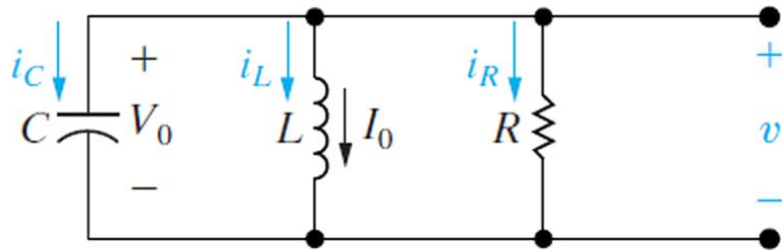


Figure 8.1 ▲ A circuit used to illustrate the natural response of a parallel *RLC* circuit.

Overdamped response ($\omega_0^2 < \alpha^2$)

$$v = A_1 e^{s_1 t} + A_2 e^{s_2 t},$$

Underdamped response ($\omega_0^2 > \alpha^2$)

$$v(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t,$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}.$$

ω_d : damped frequency,
 α : damping factor

✓ Smaller dissipative losses: the oscillations persist longer.

Critically damped response ($\omega_0^2 = \alpha^2$)

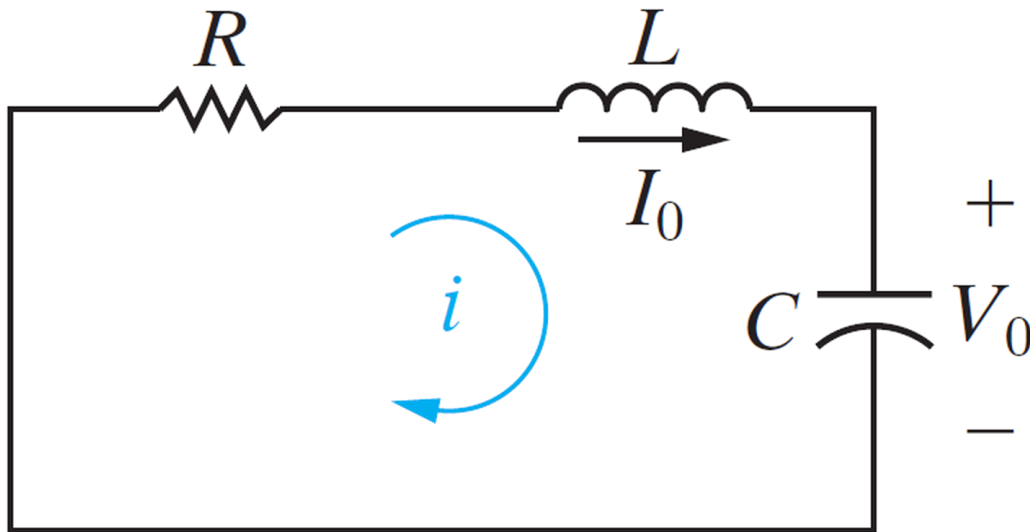
✓ Approach as quickly as possible toward equilibrium without oscillating.

$$s_1 = s_2 = -\alpha = -\frac{1}{2RC}$$

$$v(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}.$$

Series RLC Natural Response

$$Ri + L\frac{di}{dt} + \frac{1}{C} \int_0^t i d\tau + V_0 = 0$$



SERIES RLC NATURAL-RESPONSE FORMS

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \text{ (overdamped),}$$

$$i(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t \text{ (underdamped),}$$

$$i(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t} \text{ (critically damped).}$$

Homework

Rizzoni, Principles and Applications of Electrical Engineering, 7th ed., Chapter 4

- 4.23, 4.30
 - initial conditions and the Thévenin time constant (Section 4.2)
- 4.37, 4.39, 4.40, 4.47
 - first-order RC and RL transients (Section 4.3)
- 4.61, 4.75
 - second-order circuits: final values and natural response (Section 4.4)